ASSOCIATED DETAIL SPECIFICATION

RESISTORS, FIXED, FILM, HIGH VOLTAGE,
STYLES RHV22, RHV23, RHV24, RHV25,
RHV26, RHV27, RHV28, RHV29

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RHV22, RHV23, RHV24, RHV25, RHV26, RHV27, RHV28, and RHV29 resistors.

1.2 Military part number. Resistors covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:



The coded number shall be derived in accordance with MIL-R-49462.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-39032 - Resistors, Packaging of.

MIL-R-49462 - Resistors, Fixed, Film, High Voltage, General Specification for.

(Copies of the specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-49462, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be as specified in table I, based on a full load operation at an ambient temperature of 125°C. For operation at temperatures higher than 125°C, derate in accordance with figure 2.

TABLE I. Power rating.

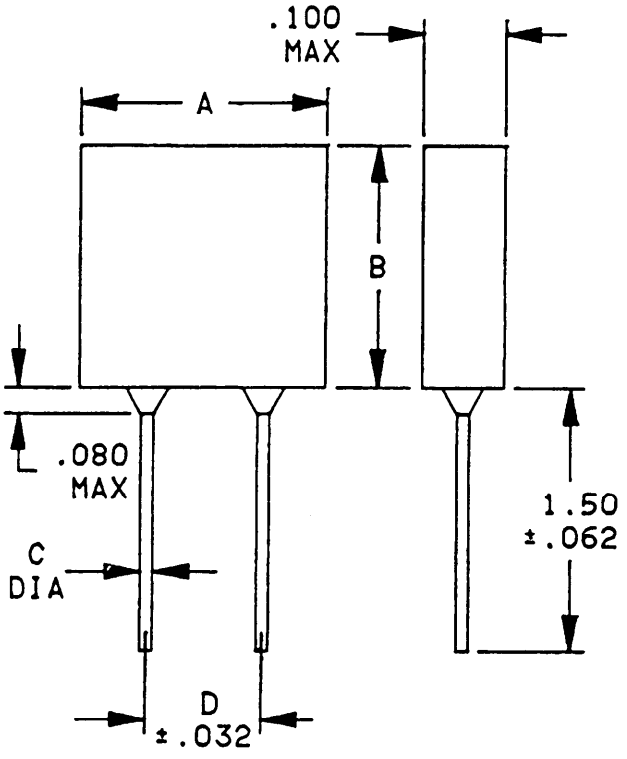
Resistor style	Power rating (watts)	Style
RHV22	0.5	A
RHV23	1.6	B
RHV24	4.0	C
RHV25	5.0	D
RHV26	0.25	E
RHV27	0.8	F
RHV28	1.5	G
RHV29	2.0	H

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table II.

TABLE II. Minimum and maximum nominal resistance values.

Resistor style	Resistance value (ohms)		Style
	Minimum	Maximum	
RHV22	10k	499M	A
RHV23	10k	2G	B
RHV24	20k	10G	C
RHV25	30k	10G	D
RHV26	100	10G	E
RHV27	100	2G	F
RHV28	100	4.9G	G
RHV29	100	8.06G	H

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Inches	mm
.005	0.13
.020	0.51
.032	0.81
.062	1.58
.080	2.03
.100	2.54
1.500	38.10

FIGURE 1. Resistor, fixed, film, high voltage.

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Dimensions								
	RHV22	RHV23	RHV24	RHV25	RHV26	RHV27	RHV28	RHV29
A	0.350 (8.9)	0.520 (13.2)	1.020 (25.9)	2.070 (52.6)	0.570 (14.5)	1.070 (27.2)	1.570 (39.9)	2.070 (52.6)
B max	0.350 (8.9)	0.520 (13.2)	0.520 (13.2)	0.750 (19.1)	0.210 (5.3)	0.280 (7.1)	0.330 (8.4)	0.330 (8.4)
C	.020 ±.005 (.508 ±1.27)	.020 ±.005 (.508 ±1.27)	.020 ±.005 (.508 ±1.27)	.032 ±.002	.020 ±.005 .508 ±.127	.020 ±.005 .508 ±.127	.032 ±.002 (.81 ±.05)	.032 ±.002 (.81 ±.05)
D	0.200 (5.1)	.400 (10.16)	.900 (22.86)	1.900 (48.26)	.400 10.16	.900 22.86	1.400 35.56	1.900 48.26

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.

FIGURE 1. Resistors, fixed film, high voltage - Continued.

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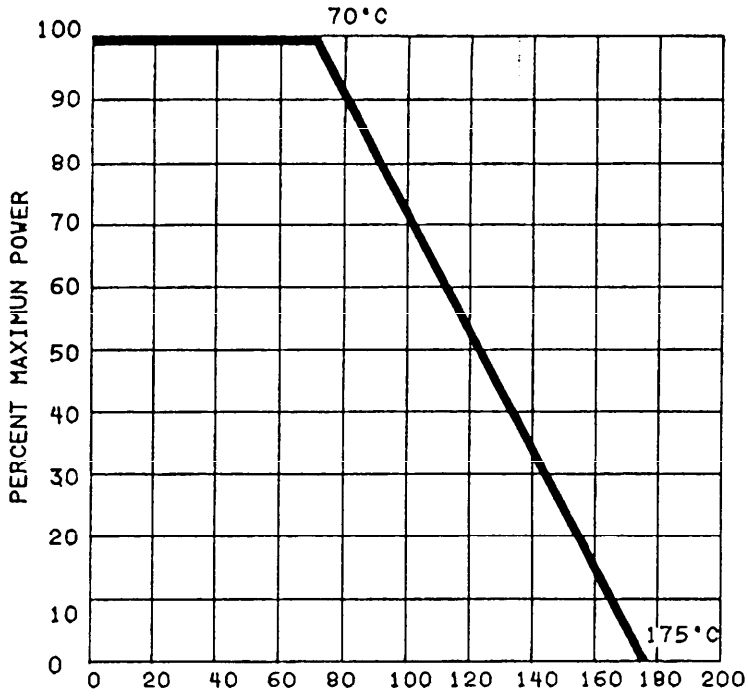


FIGURE 2. Derating curve.

3.5 Maximum voltages. Maximum voltages shall be in accordance with table III.

TABLE III. Maximum voltages.

Resistor style	Voltage (volts maximum) (kV)	Style
RHV22	1.5	A
RHV23	3.5	B
RHV24	7.5	C
RHV25	25.0	D
RHV26	3.75	E
RHV27	7.5	F
RHV28	11.25	G
RHV29	15.0	H

3.6 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistance tolerance percent (±)
F	1.0
G	2.0
J	5.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling inspection. Sampling and inspection shall be in accordance with MIL-R-49462.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

6.1 Intended use. Resistors are intended for use in electronic circuits where high voltages and high resistances are required.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1063-02)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-49462/2	RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV22 through RHV29

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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